

# AI Sales Forecasting Dashboard using SARIMA, Prophet, and XGBoost

## Submitted By

**Name:** Ketan Mahakal

**Degree:** Bachelor of Engineering (Information Technology)

**Institute:** L. D. College of Engineering, Ahmedabad

**Academic Year:** 2026

## Abstract

Sales forecasting plays an important role in business planning by helping organizations estimate future demand, optimize inventory, and improve strategic decision-making. In this project, a sales forecasting system was developed using historical Superstore sales data. The project involved data preprocessing, exploratory data analysis, time series forecasting, and the development of an interactive dashboard.

Three forecasting techniques—SARIMA, Prophet, and XGBoost—were implemented and compared using MAE, RMSE, and MAPE. The final solution also includes a Streamlit dashboard that enables users to explore sales trends, compare forecasting models, and view business insights in an interactive way.

This project demonstrates practical applications of machine learning and time series analysis for solving real-world business problems.

## Chapter 1: Introduction

Sales forecasting is one of the most important analytical tasks for retail and e-commerce businesses. Accurate forecasts help organizations plan inventory, manage supply chains, allocate resources, and reduce operational costs.

Many businesses still rely on manual estimation or simple historical averages, which often fail to capture seasonal patterns and changing customer demand.

The objective of this project is to build a data-driven forecasting system capable of predicting future monthly sales using historical transaction data. Along with forecasting, the project provides visual analysis and business insights through an interactive dashboard.

## Chapter 2: Problem Statement

Retail businesses often face challenges such as:

- Uncertain customer demand
- Overstocking and understocking
- Inefficient inventory planning
- Poor demand estimation
- Revenue loss due to inaccurate forecasts

To address these issues, this project develops an intelligent forecasting system that analyzes historical sales data and predicts future sales using multiple forecasting algorithms.

**Chapter 3: Objectives**

The primary objectives of this project are:

- To clean and preprocess sales data.
- To perform exploratory data analysis.
- To identify sales trends and seasonal patterns.
- To build forecasting models using SARIMA, Prophet, and XGBoost.
- To compare model performance using standard evaluation metrics.
- To develop an interactive dashboard using Streamlit.
- To provide actionable business insights based on forecasting results.

**Chapter 4: Dataset Description**

The project uses the Superstore Sales Dataset, which contains retail sales transactions from multiple regions and product categories.

**Dataset Information**

| Attribute    | Description          |
|--------------|----------------------|
| Order Date   | Date of purchase     |
| Ship Date    | Shipping date        |
| Category     | Product category     |
| Sub-Category | Product sub-category |
| Region       | Sales region         |
| Segment      | Customer segment     |
| Sales        | Sales amount         |

The dataset contains approximately 9,800 records and 18 attributes covering multiple years of sales transactions.

**Chapter 5: Data Preprocessing**

Before analysis, the dataset was cleaned and prepared to improve data quality.

The preprocessing steps included:

- Converting date columns into datetime format.
- Handling invalid date values.
- Removing duplicate records.
- Checking missing values.
- Creating monthly sales data for forecasting.
- Organizing data into categories and regions.

These steps ensured that the forecasting models received consistent and reliable input data.

## **Chapter 6: Exploratory Data Analysis (EDA)**

Exploratory Data Analysis was performed to understand historical sales behavior.

The following analyses were conducted:

### **Monthly Sales Trend**

Monthly sales were aggregated to observe long-term trends and seasonality.

### **Category Analysis**

Sales were compared across product categories to identify high-performing products.

### **Region Analysis**

Regional sales performance was evaluated to understand geographical demand.

### **Sub-Category Analysis**

Detailed analysis was carried out for individual product groups.

### **Sales Distribution**

Histograms and box plots were used to examine the distribution of sales values and detect outliers.

## **Chapter 7: Forecasting Models**

Three different forecasting techniques were implemented.

### **7.1 SARIMA**

SARIMA is a statistical time series forecasting model designed to capture seasonal behavior.

Advantages:

- Handles seasonality
- Suitable for historical monthly data
- Widely used for business forecasting

## 7.2 Prophet

Prophet is an open-source forecasting library developed by Meta.

Advantages:

- Automatic trend detection
- Handles seasonality effectively
- Works well with business time series
- Easy to tune and interpret

## 7.3 XGBoost

XGBoost is a machine learning algorithm based on gradient boosting.

Advantages:

- Captures nonlinear relationships
- High prediction accuracy
- Efficient training
- Strong performance on structured datasets

## Chapter 8: Model Evaluation

The forecasting models were evaluated using three standard metrics.

### Mean Absolute Error (MAE)

Measures the average prediction error.

### Root Mean Squared Error (RMSE)

Penalizes larger forecasting errors and is commonly used for regression evaluation.

### Mean Absolute Percentage Error (MAPE)

Represents forecasting error as a percentage, making model comparison easier.

The model with the lowest error values was selected as the best forecasting model.

## Chapter 9: Dashboard Development

An interactive dashboard was developed using Streamlit.

The dashboard includes:

- Home Dashboard

- Sales Analysis
- Category Analysis
- Region Analysis
- Forecasting Results
- Model Comparison
- Business Insights

The dashboard allows users to interact with sales data and visualize forecasting results without writing any code.

### **Chapter 10: Business Insights**

The analysis revealed several useful business observations.

- Technology products generated the highest sales.
- The West region contributed the largest share of revenue.
- Monthly sales followed clear seasonal trends.
- Forecasting models successfully predicted future monthly sales.
- XGBoost demonstrated strong forecasting performance on the available dataset.

These findings can help businesses improve inventory planning, demand forecasting, and marketing strategies.

### **Chapter 11: Challenges Faced**

During development, several challenges were encountered.

- Date format inconsistencies.
- Time series preprocessing.
- Hyperparameter tuning.
- Prophet installation and configuration.
- Comparing different forecasting models.
- Integrating machine learning outputs into the Streamlit dashboard.

Each challenge was addressed through careful preprocessing, debugging, and model evaluation.

### **Chapter 12: Conclusion**

This project successfully developed an end-to-end sales forecasting system using statistical and machine learning techniques.

Historical sales data was analyzed to identify trends and seasonal patterns. Multiple forecasting models were implemented and evaluated using standard performance metrics. A Streamlit dashboard was created to present forecasting results and business insights through an interactive interface.

The project demonstrates how data analytics and machine learning can support business decision-making and demand planning.

### **Chapter 13: Future Scope**

The project can be extended in several ways.

- Real-time sales forecasting
- LSTM and Transformer-based forecasting models
- Customer demand prediction
- Product recommendation systems
- Cloud deployment
- Integration with Power BI
- Live sales data using APIs

### **Technologies Used**

- Python
- Pandas
- NumPy
- Matplotlib
- Plotly
- Streamlit
- Prophet
- XGBoost
- Statsmodels
- Scikit-learn
- Joblib

### **Project Outcomes**

- Cleaned and processed retail sales data.
- Performed exploratory data analysis.
- Built three forecasting models.
- Compared model performance using MAE, RMSE, and MAPE.
- Developed an interactive Streamlit dashboard.
- Generated business insights for sales planning.

## **References**

1. Superstore Sales Dataset
2. Python Documentation
3. Streamlit Documentation
4. Prophet Documentation
5. XGBoost Documentation
6. Scikit-learn Documentation
7. Statsmodels Documentation